

Derbyshire Shared Care Pathology Guidelines

Hyperthyroidism in Adults.

Purpose of Guideline

The investigation, management and referral criteria of adult patients with newly diagnosed hyperthyroidism.

Definition

Suppressed TSH and raised Free T4 &/or Free T3.

Background

- Hyperthyroidism has a significant short-term morbidity and long-term morbidity and mortality
- The most common causes of hyperthyroidism in iodine-replete communities are autoimmune Graves' disease, toxic multinodular goitre and excessive thyroxine replacement
- Other causes include autonomously functioning thyroid adenoma and thyroiditis
- If there are features of thyroiditis (recent virus, neck tenderness, absence of primary thyroid disease signs), avoid anti-thyroid drugs, prescribe B-blockers if no contra-indications and seek Endocrine advice dependent on severity of symptoms, especially if TRAb negative
- The prevalence of hyperthyroidism in women is 0.5-2% but is only a tenth of that in men

When is hyperthyroidism suspected?

Symptoms include:

- Weight loss
- Tachycardia
- Weakness and tiredness
- Sweating
- Irregular periods/ infertility

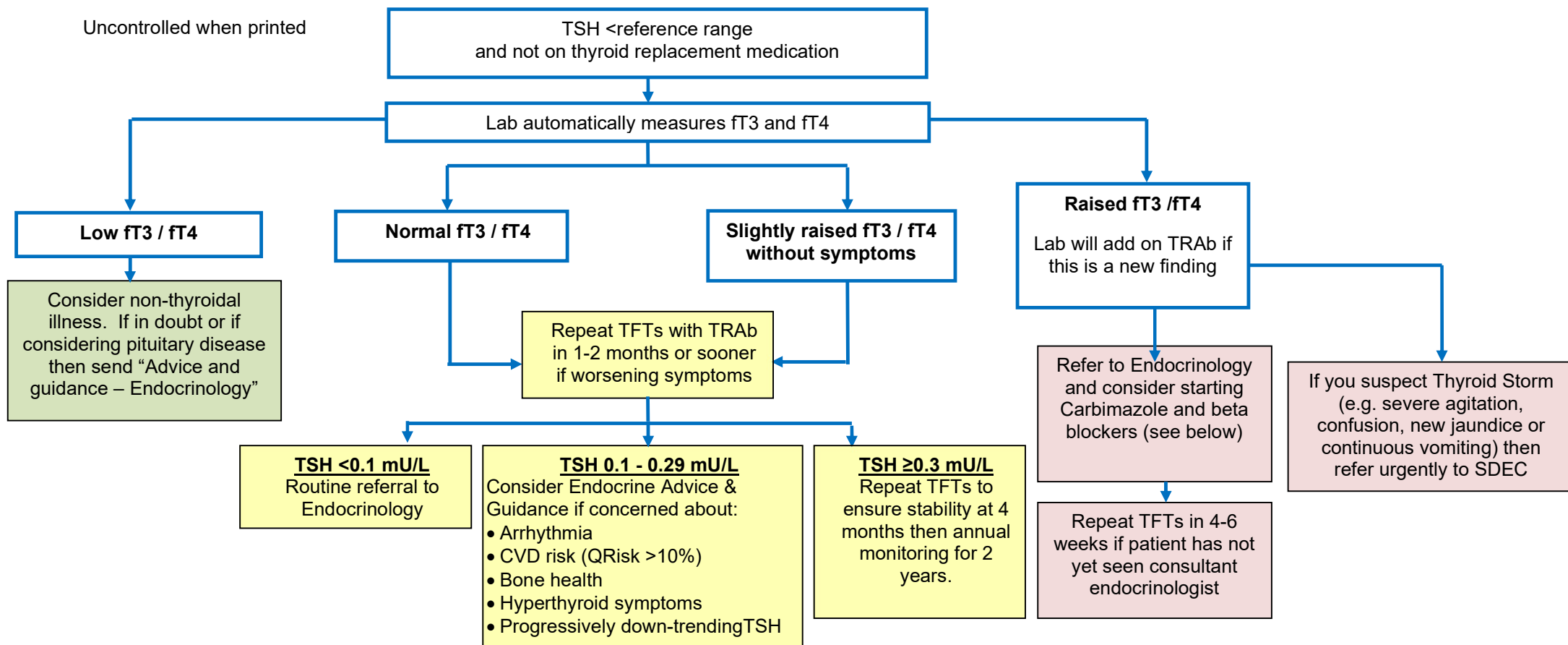
The number and severity of symptoms will be dependent on the degree of hyperthyroidism. Patients may be asymptomatic especially if they have subclinical hyperthyroidism.

When hyperthyroidism is considered a medical emergency:

There are no absolute values for Free T4/T3 which can indicate whether a patient with hyperthyroidism should be considered as a medical emergency. Other factors will also determine whether the patient requires emergency admission, including:

- End-Organ failure e.g. kidney, liver or heart failure
- Pyrexia
- Atrial Fibrillation with fast ventricular response
- Anti-thyroid drug induced liver failure

Please provide any relevant clinical or medication details with the request (e.g. dose of thyroxine, liothyronine, carbimazole, propylthiouracil, or if patient is on drugs that are known to affect TFTs (e.g. Amiodarone, lithium, immunotherapy for cancer). This will enable appropriate interpretation and help the laboratory to gauge whether this patient's results need to be telephoned urgently or not.



- If TRAb Negative AND other symptoms of thyroiditis are present repeat TFT in 4-6 weeks PRIOR to referral

Carbimazole

- If Pregnant, Carbimazole should not be started in primary care without discussion with Endocrinology.
- If Carbimazole is started:
 - Please ensure referral to Endocrinology. Advice to females about contraception should be considered. Advice should be given about reporting signs or symptoms suggestive of infection (especially sore throat) immediately to a doctor
 - **Side effects** include a rash, arthralgia or neutropaenia
If the patient develops a severe sore throat, severe mouth ulcers and/or fever, please check full blood count and stop carbimazole until WBC confirmed as normal. If neutropaenic refer to Haematologists and stop anti-thyroid medication.
 - β -blockers (e.g. Propranolol, Bisoprolol) can provide symptomatic relief whilst TFTs normalise
 - **Dose**
 - If $fT4 > 30$ pmol/L or $fT3 > 10$ pmol/L then start Carbimazole 20mg once a day
 - If $fT4$ 22-30 pmol/L or $fT3$ 6.9-9.9 pmol/L then start Carbimazole 10mg once a day

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TSH Receptor Antibodies (TRAb)

TSH receptor antibodies are autoantibodies directed against the TSH receptor. Stimulatory TRAb's are associated with Graves' disease and measurement is recommended by NICE (NG145) to help delineate between Graves' disease (and its clinical sequelae) and other causes of hyperthyroidism.

Thyrotoxicosis and pregnancy

- Pregnant women with current thyrotoxicosis on anti-thyroid medication should be **urgently referred** to endocrine antenatal clinic
- Pregnant women with a past history of medically, surgically or radioiodine-treated hyperthyroidism who are currently euthyroid should be referred to endocrine antenatal clinic on a non-urgent basis

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am to 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Burton

Duty Biochemist	01283 593313 (8am - 4 pm, Mon-Fri)
On Call Consultant Biochemist	Via QHB switchboard, 01283 566333
Endocrinology Advice (urgent)	Via QHB switchboard, 01283 566333
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

References

1. Thyroid disease: assessment and management. NICE guideline NG145 (Nov 2019, updated Oct 2023). <https://www.nice.org.uk/guidance/ng145>
2. UK Guidelines for Thyroid Function Tests (ACB, British Thyroid Association and British Thyroid Foundation, 2006). https://www.british-thyroid-association.org/sandbox/bta2016/uk_guidelines_for_the_use_of_thyroid_function_tests.pdf

Patient information sheets

Patient information sheets can be obtained from the British Thyroid Foundation and Patient.co.uk.

<http://www.btf-thyroid.org/information/leaflets/41-hyperthyroidism-guide>

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<http://www.patient.co.uk/health/Hyperthyroidism-Overactive-Thyroid.htm>

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	V2 Sept 2013	Dr R Rea, Dr J Monaghan, Dr P Blackwell,	Routine Review
	V3 Oct 2015	Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Routine Review
	V4 Nov 2017	Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Routine review
	V5 Jan 2021	Dr D Hughes, Dr H Ali, Dr R MacInerney, Dr P Shillo, Dr P Blackwell, Mrs H Seddon	Routine review
	V6 Sept 2023	Dr H Ali, Dr A Ugur, Dr R Robinson, Dr P Blackwell, Mrs H Seddon	Routine review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	12/11/2025	Dr A Ugur	Consultant Endocrinologist
Secondary Care CRH	04/11/2025	Dr P Shillo	Consultant Endocrinologist
Primary Care	13/11/2025	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	13/11/2025	Ms S Knowles	Consultant Clinical Scientist